Credit Card Analytics Dashboard

Customer & Transaction Analysis

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Tools Used: MS Power BI & MS Excel

Project Description:

This project analyzes credit card customer demographics and transaction behavior to identify key revenue drivers, customer segments, and spending patterns that can help financial institutions improve targeting and profitability.

Business Problem & Objective

Business Problem

Banks generate large volumes of customer and transaction data but struggle to:

- Identify high-value customers
- Understand spending behavior
- Optimize card offerings and marketing strategies

Project Objectives

- Analyze customer demographics and income segments
- Identify top revenue contributors
- Understand spending categories and payment behavior
- Provide actionable business insights for decision-making

Dataset Overview

Customer Data

- Age, Gender, Income Group
- Job Type, Education
- Number of Dependents

Transaction Data

- Transaction Amount
- Card Category
- Payment Mode
- Expenditure Type
- Time (Week/Quarter)

Key Performance Indicators

Total Revenue

221M

Total Transaction Amount

178.1M

Total Interest Earned

31M

Total Transactions

2.6M

Dashboard Overview

Features

- **Interactive filters:** Week, Gender, Quarter, Card Type
- **Two analytical views:**
  1. Customer Insights
  2. Transaction Insights
- Real-time segmentation and performance tracking

Value

Enables business users to explore revenue trends and customer behavior dynamically.

Customer Insights

Revenue by Age

- Age group **40–50** generates the highest revenue (97M)
- Followed by 50–60 segment (73M)

Revenue by Income

- High-income customers contribute **117M**
- Middle-income: 63M
- Low-income: 41M

Revenue by Job Type

- Businessman: 70M
- White-collar: 40M
- Self-employed and Government follow

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Insight: Middle-aged, high-income professionals are the most valuable customer segment.

Demographic & Geographic Trends

Education Level Contribution

- Graduates: **89M** (highest)
- High School: 44M
- Others contribute moderate revenue

Dependent Analysis

- Customers with **2–3 dependents** show higher spending patterns

Top Performing States

- Texas (51M)
- New York (51M)
- California (50M)

Insight: Educated working professionals in major states drive the majority of revenue.

Transaction & Spending Analysis

Revenue by Card Type

- Blue Card: **185M** (major contributor)
- Silver: 22M
- Gold: 10M
- Platinum: 5M

Quarterly Trend

- Highest revenue in **Q3 (57M)**
- Slight decline in Q4

Expenditure Categories

- Bills: 55M
- Entertainment: 38M
- Fuel: 37M
- Grocery & Food follow

Payment Mode

- Swipe: 140M
- Chip: 68M
- Online: 14M

Insight: Essential expenses dominate, and physical transactions are still preferred.

Key Insights, Recommendations & Conclusion

Key Insights

- Revenue is driven by middle-aged, high-income professionals
- Blue card users form the largest revenue base
- Spending is concentrated on essential categories
- Digital payment adoption is relatively low

Business Recommendations

- Upsell Gold/Platinum cards to Blue card users
- Target 40–50 age group with personalized offers
- Promote online payment incentives
- Focus marketing in high-performing states

Conclusion

The dashboard enables data-driven decision-making by identifying high-value customers, spending patterns, and growth opportunities for the credit card business.